Higher National Diploma in Information Technology
Second Year, Second Semester Examination – 2021
HNDIT2413-Web Application Development

Instructions for Candidates:	No. of questions : 06
Answer five (5) questions only.	No. of pages : 07
All questions carry equal marks	Time : Three hours(03)

01.

i. What do you mean by web Application? [02 Marks]

ii. Briefly explain a web browser and a search engine with an example. [04 Marks]

iii. Write the html code to create the following list. [06 Marks]

Learning Web Development

1. Background Skills
   A. Unix Command
   B. Vim Text Editor
2. HTML
   A. Minimal Page
   B. Headings
   C. Tags
   D. Lists
      I. Unordered
      II. Ordered
      III. Definition
      IV. Nested
   E. Links

iv. Write the output of the following HTML code. [08 Marks]

```
<!doctype html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>tableSpan</title>
</head>
<body>
 <table border = "1">
 <caption><p>My Daily Routine</p></caption>
 <tr>
 <th></th>
 <th>Monday</th>
 <th>Tuesday</th>
 <th>Wednesday</th>
 <th>Thursday</th>
 <th>Friday</th>
 </tr>
 <tr>
 <th>Breakfast</th>
 <td>Chappathi</td>
 <td>Vegetable Salad</td>
 <td>Mango Juice</td>
 <td>Dosai</td>
 <td>Idly</td>
 </tr>
 <tr>
 <th>Morning Workout</th>
 <td colspan = "2" align = "center">Walking 30 mins</td>
 <td colspan = "3" align = "center">Cycling 30 mins</td>
 </tr>
 <tr>
 <th>Lunch</th>
 <td>Rice and Curry</td>
<td>Kottu</td>
 <td>Biriyani</td>
 <td>Fried Rice</td>
 <td rowspan = "2">Vegetable Mix</td>
 </tr>
 <tr>
 <th>Dinner</th>
 <td>Bread</td>
 <td>Fruits</td>
 <td>Soup</td>
 <td>Grains</td>
 </tr>
 </table>
</body>
</html>
```

[Total 20 Marks]

02.

i. How do you insert an external CSS File named as **heading.css** in to a web page? Write down the appropriate HTML tag and attributes. [02 Marks]

ii. Write two (02) differences between external and inline CSS. [04 Marks]

iii. State three (03) benefits of using CSS for a web site. [06 Marks]

iv. Consider the below output and fill the blanks in the given internal stylesheet. [08 Marks]

Introducing ChatGPT

ChatGPT is an artificial intelligence (AI) tool that can answer questions, tell stories, produce essays and even write code.

The chatbot is so sophisticated that people are suddenly starting to see the true potential of AI and how, for better or for worse, it could define humanity's future.

Developed by OpenAI in San Francisco, a free version of ChatGPT is available for anyone to use on the ChatGPT website. All you have to do is sign up to get a login, and you can be mining the depth of the AI model in seconds.

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```
<!DOCTYPE html>
<html>
<head>
<style>
body {
        background:url("thayro.png") no-repeat fixed center;
      }
h1{
  color: Darkblue;
  margin-left: 70px;
  margin-right: 50px;
  border: 10px 1)……................;
  }
p.left {
        text-align: left;
        font-family: Helvetica;
        2)……..……: 16pt;
      }
p#center {
        text-align: 3)……..……..;
        font-size: 16pt;
        font-family:Georgia;
        }
```

```
div.footer {
        font-size: 15pt;
        text-align: right;
          }
4)..........................
</head>
<body>

<h1>Introducing ChatGPT</h1>
<p 5)................>ChatGPT is an artificial intelligence (AI) tool that can answer questions, tell stories, produce essays and even write code.</p>
<p 6)...................>The chatbot is so sophisticated that people are suddenly starting to see the true potential of AI and how, for better or for worse, it could define humanity's future. </p>
<p id="center">Developed by OpenAI in San Francisco, a free version of ChatGPT is available for anyone to use on the ChatGPT website. All you have to do is sign up to get a login, and you can be mining the depth of the AI model in seconds.</p>
<div 7)..........................>
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8)..........................
</body>
</html>
```

[Total 20 Marks]

03.

i. Differentiate the client-side script and server-side script. [02 Marks]

ii. Mention two uses of client-side scripting languages. [04 Marks]

iii. Write the output of the following code snippet in javascript. [06 Marks]

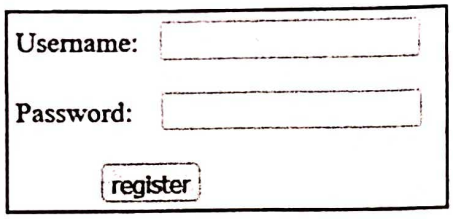
```
<script language="javascript" type="text/javascript">
    names = new Array("Pravin Raagha", "Thayro TDY");
    var y=0;
    var student="Vikum";
    document.write(names[0].substring(0,6));
    document.write(names[1].substring(6));
    document.write("<br>");
    document.write(names[1].charAt(5));
    document.write("<br>");
    for(var x=0;x<=5;x++)
    {
    y=Number(y)+Number(x);
    }

    document.write("Answer is : "+y);
    greetings=(student=="VIKUM")?"Friend  ":"Dear ";
    document.write("<br>"+"Have a good day "+greetings);
</script>
```

iv.

a. Write a html code to display the following form (form name is **myform**)

[02 Marks]



b. Write a JavaScript function to validate the username and password. The username can't be empty and password can't be less than 6 characters long. (Username textbox's name is **uname** , Password textbox's name is **password**)

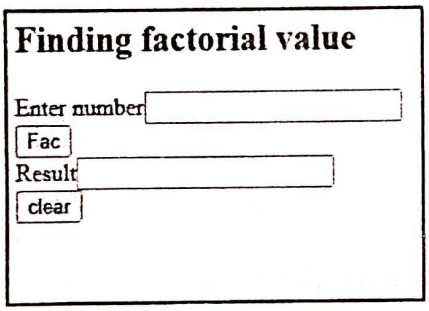
[06 Marks]

[Total 20 Marks]

04.

i. What do you mean by function in JavaScript? [02 Marks]

ii. Write a html program to display the following form. [04 Marks]



iii. The above form is used to find the factorial value for the given number.

a. When you click the "fac" button, factorial value should be displayed on "result" text box. Write a function for that.
(eg. Factorial 4,4!=4*3*2*1=24)

[07 Marks]

b. When you click the "Clear" button, form should be cleared. Write a function for that.

[03 Marks]

iv. Write the output of following program. [04 Marks]

```
<html>
<head>
<script language="JavaScript" type="text/JavaScript">
function Land_Area() {
return this.width*this.length;
}
```

```
function Land_Perimeter() {
return 2*(this.width+this.length);
}
function Land(w,l)
{
this.width=w;
this.length=l;
this.area=Land_Area;
this.perimeter=Land_Perimeter;
}
myland=new Land(5,4);
document.write("Output is: ");
document.write("<br>");
document.write(myland.width);
document.write("<br>");
document.write(myland.perimeter());
document.write("<br>");
document.write(myland.area());
</script>
</head>
<body>
</body>
</html>
```

[Total 20 Marks]

05.

i. Define cookies and Session. [04 Marks]

ii. Write a PHP program to keep two students' details (Name, NIC, Age, Address) and display it. (**Hint** : use multidimensional array) [08 Marks]

iii. Write PHP code segments to do the following tasks.

a. Hold five numeric values as single variable using this numbers 8,9,3,7 and10

b. Find 12 is in the array

c. Find the maximum value of the above elements.

d. Sort the values in **descending** order of the above elements. [08 Marks]

[Total 20 Marks]

06. Answer the below questions based on the given instructions.

DBMS : MySQL

Web Programming Language : PHP

Database Server : localhost

User Name : root

Password : rc123

Database Name : YGC

Table Name: inventor

Write PHP code fragment to do the following.

i. Write PHP code segment to create the connection and check the connection to the MYSQL database. [03 Marks]

ii. Write PHP code segment to select the database "YGC" and create the table inventor. [05 Marks]

[**Hint**: table structure-
inventor (Invid,Firstname, Lastname, InventionCategory, Age)
Invid is a primary key]

iii. Write PHP code segment to insert the below data set into the inventor table. [04 Marks]

Firstname: Prof.Thani
Lastname: Darun
InventionCategory: Engineering
Age:32

iv. Write PHP code segment to change the Lastname to Krishz of the inventor whose Invid=1 and display the record of the updated inventor as follows.

[08 Marks]

Invid:1
Firstname: Prof.Thani
Lastname: Krishz
InventionCategory: Engineering
Age:32

[Total 20 Marks]